CONTINUED

Data can also be stolen. This is an additional threat if the data is stored online, e.g. on a web server. This is because external people can access the web server and access the data on it. This could include personal information such as your bank details. This is why it is important that the web servers have security such as firewalls to attempt to stop this unauthorised access. You can learn more about online security in Chapter 5.

Discussion questions

1 Consider a social media website. What information will they store about their users? What could happen if someone gains unauthorised access to this data?

2 How important is it that a company keeps their data secure? What are the potential consequences of someone gaining unauthorised access to personal data?

9.1 Database structure

Introduction to databases

A **database** is a collection of data that is set up in a structured way. This is usually on a computer, but you could have a paper-based database as long as it is suitably structured.

A database holds data about objects, for example, a person, an item in a shop, an order someone has made, etc. Each of these objects is stored in a **table**.

Each table is then made up of **fields**, a field is an individual piece of data being stored about an object. For a person you could have a field for first name, a field for email address, for phone number, etc.

A collection of these fields about one object is called a **record**. For example, all the information about the person Aarna Singh will be one record.

Most databases have many tables and these are all joined together so that the data is linked. This is outside the scope of the specification. You only need to look at single table databases.

Here's an example database about books in a library. The table is called `books`.

Book name	Author	Publisher	Genre	Fiction
Picking daisies	J. Frank	Cambridge horticulture	Gardening	False
Night stars	K. Mars	Si-fi books	Science fiction	True
Dreaming of the sun	P. Yu	Si-fi books	Science fiction	True
Cooking for fun	W. Crisp	Cookery penguin	Cookery	False

This database has five fields; Book name, Author, Publisher, Genre and Fiction.

This database as four records: One for Picking daisies, one for Night starts, one for Dreaming of the sun and one for Cooking for Fun.

KEY WORDS

database: an example of application software to store and manipulate data.

table: a set of data about one type of object, e.g. students.

field: an individual piece of data, e.g. date of birth.

record: all of the records in a table about one object, e.g. all the personal details about one student.

ACTIVITY 9.1

Consider your school. In pairs identify the fields that your school will store about you. Write a list of the fields. Compare your pair's answers with another pair. Were there any differences? Did you add any more to your list?

Data types

Each field has a specific **data type**. This identifies what that field can hold. These are similar to the data types you learnt about in programming (Chapter 8).

Table 9.1 shows the common database data types:

Data type	Description	Examples
Text Alphanumeric	Any combination of letters, symbols and/or numbers. The numbers are treated as text and not numeric, i.e. you can't add them together. These are always speech marks around the data.	"Hello" "2JK8D" "2198" "Hi!!"
Character	A single letter, symbol or number. If it is a number it will be treated as text, i.e. you cannot add to it.	"H" "y" "1" "?"
Boolean	One of two options usually True / False or Yes / No. The data does not have speech marks around it.	True False
Integer	A whole number.	123 999 0 −1928
Real	A number with at least one decimal place.	0.0 1.2 9.99 100.92 −2.9384
Date/time	A date and/or a time.	1/1/2020 8:30 2/3/2020 16:00

Table 9.1: Database data types

KEY WORD

data type: the characteristics of a piece of data. Common data types are string, integer, real and Boolean.

KEY WORDS

Boolean: a data type. True or False.

integer: a data type. Whole numbers.

real: a data type. Decimal numbers.

Primary key

A **primary key** is a unique field in a database. This means that it cannot appear twice in different records. It is used to uniquely identify a record. For example, if you want to find a specific person in a database and you enter their name, there could be dozens of people with the same name. If they had an ID number that is unique to only them, then you can access their record straight away.

KEY WORD

primary key: a unique identifier for a record.

Look at this database again:

Book Name	Author	Publisher	Genre	Fiction
Picking daisies	J. Frank	Cambridge horticulture	Gardening	False
Night stars	K. Mars	Si-fi books	Science fiction	True
Dreaming of the sun	P. Yu	Si-fi books	Science fiction	True
Cooking for fun	W. Crisp	Cookery penguin	Cookery	False

Could any of these five fields be unique?

- Book name: there could be two books with the same names.
- Author: one author could write more than one book.
- Publisher: one publisher could publisher more than one book.
- Genre: more than one book could have the same genre.
- Fiction: more than one book could be False, and more than one could be True.

This means that none of these fields is a suitable primary key. So instead we add a key field:

Book ID	BookName	Author	Publisher	Genre	Fiction
1	Picking daisies	J. Frank	Cambridge horticulture	Gardening	False
2	Night stars	K. Mars	Si-fi books	Science fiction	True
3	Dreaming of the sun	P. Yu	Si-fi books	Science fiction	True
4	Cooking for fun	W. Crisp	Cookery penguin	Cookery	False

The field Book ID becomes the primary key.

Defining a single-table database

To define a single-table database you will need to decide:

1. the fields that you need
2. the data types for each field
3. the primary key.

Example:

> A shop that sells wool needs to store the items it has in stock. Each item has; a name, e.g. "Sparkle wool", a colour, e.g. "red", a weight, e.g. 2, a price, e.g. $3.99 and a quantity in stock, e.g. 23.

To define a single-table database for the shop, you need to:

1. Identify the fields that you need

Read the description again and for each item it describes select a field name.

Description	Field name
Each item has; a name, e.g. "Sparkle"	Item name
a colour, e.g. "red"	Colour
a weight, e.g. 2	Weight
a price, e.g. $3.99	Price
a quantity in stock, e.g. 23.	Quantity

2. Identify the data types for each field

Look at the example data and the description and select the most appropriate data type from the table.

Field name	Explanation	Data type
ItemName	"Sparkle" – this is letters therefore string or character. There is more than one letter so it is not character.	Text
Colour	"red" this is letters and there is more than 1 character.	Text
Weight	2 – this is numeric, there is no decimal so it is not real.	Integer
Price	$3.99 – this is numeric, there is a decimal so it is not integer.	Real
Quantity	23 – this is numeric, there is no decimal so it is not real.	Integer

3. Identify a primary key

None of these fields are unique, you could have two with the same name, same colour, etc. Therefore a new ID field needs adding to become the primary key.

Field name	Data type
IDNumber	Text
ItemName	Text
Colour	Text
Weight	Integer
Price	Real
Quantity	Integer

COMPUTER SCIENCE IN CONTEXT

Every time someone creates a new database they have to go through the same process you have just been through. Consider a library. When they set up a new database they have to program the database with the fields and data types that they need to use.

PROGRAMMING TASK 9.1

A clothes store needs a single-table database to store data about its stock. It needs to store the barcode of each item, the name of the product, the colour, size, quantity in stock and whether an order for more products has been made or not.

Getting started

1 Explain what is meant by a record and a field in a database.

2 Identify **four** data types and describe the data they store.

3 Identify what is meant by a primary key.

Practice

1 Identify the fields for your database table.

2 Identify the data types for your fields.

3 Identify the primary key for your table.

4 Create a paper-based table for this system and complete several records.

Challenge

This task goes beyond the range of the specification.

Open a database software, e.g. Microsoft Access, and find out how to create a table and add data to it. Create the database for the shop.

Questions

1 Complete the table by giving an example of each type of data.

Data type	Example
Boolean	
Integer	
Text	
Character	
Real	
Date/time	

2 Describe the format of a database table.

3 Identify the purpose of a primary key.

4 a A social media website stores the following information about its users: username (e.g. daisy123), password (e.g. HWHRU2D), online name (e.g. Daisy), date account created (e.g. 9/9/2019), number of friends (e.g. 38).
Identify the fields and data types for a single-table database for the website.

b Identify an appropriate primary key for the table.

9.2 SQL

SQL stands for Structured Query Language. This is a standard language that is used across most types or database to perform actions such as:

- define tables
- change tables
- add data to tables
- search for data from tables (query)
- perform calculations using data from tables.

In this specification you only need to know the last two of these: search for data and perform calculations using data.

When you write some SQL code you are writing an SQL **script.** This is series of statements that are commands that are then executed and one or more values are returned.

KEY WORDS

Structured Query Language (SQL): a standard language used to define and manipulate databases.

script: a set of statements that are executed.

9.3 SELECT ... FROM

This allows you select a set of fields from a table. All of the data in each field will be returned.

This table is called Books.

Book ID	BookName	Author	Publisher	Genre	Fiction
1	Picking daisies	J. Frank	Cambridge horticulture	Gardening	False
2	Night stars	K. Mars	Si-fi books	Science fiction	True
3	Dreaming of the sun	P. Yu	Si-fi books	Science fiction	True
4	Cooking for fun	W. Crisp	Cookery penguin	Cookery	False

Select has the format:

```
SELECT field
FROM table
```